

Yunxi Lai

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Date of Birth: May 1998

Nationality: China
Gender: Female

EDUCATION

- Jul 2023–Jun 2026** **Doctor of Medicine (M.D.), Ophthalmology, Zhongshan Ophthalmic Center, Sun Yat-sen University, China**
Degree conferral expected December 2026.
Thesis: Transformation from Clinical Data to Scientific Evidence Enabled by Large Language Models in Ophthalmology.
- Sep 2020–Jun 2023** **Master of Medicine (M.M.), Ophthalmology, Zhongshan Ophthalmic Center, Sun Yat-sen University, China**
Completed alongside China's standardized residency training programme.
Thesis: Analyses of Routine Preoperative Blood Testing in Ocular Patients.
- Sep 2015–Jun 2020** **Bachelor of Medicine, Guangdong Medical University, China**

RESEARCH EXPERIENCE

- Jul 2023–Jun 2026** **[Founding Team Member] Development, Validation and Clinical Translation of an Ophthalmic Large Language Model (LLM)**
- Co-developed an ophthalmic LLM by shaping its specialty knowledge structure and evaluation framework, achieving performance comparable to GPT-4 in ophthalmic question answering.
- Led clinical study design and statistical analysis for a randomized controlled trial, demonstrating superior pre-consultation performance compared with ophthalmology residents in routine outpatient care.
- Supported deployment across multiple regions in China by coordinating AI-assisted screening and referral assessment, reaching over 1.7 million individuals.
- Co-authored three articles, contributed to grant materials, clinical translation and international dissemination, including the project's selection as a Chinese digital health case study at the 2025 World Health Assembly.
- Aug 2023–Mar 2026** **[Clinical Researcher] Prospective Pediatric Ophthalmology Cohort**
- Coordinated ethics approval, multicenter collaboration, participant recruitment, ophthalmic examinations and longitudinal data management across two institutions.
- Established and managed a prospective pediatric cohort of 668 children, with an average of seven follow-up visits per participant.
- Built an LLM-assisted information extraction workflow to convert clinical narratives into structured research variables with active data quality control.
- Nov 2023–Mar 2025 Intermittent** **[Licensed Ophthalmologist] Community Eye-Care Screening Cohort**
- Recruited participants, reviewed AI-generated ophthalmic reports, assessed referral recommendations, and collaborated with community healthcare providers.
- Sep 2021–Jun 2023** **[Independent Project Lead] Routine Preoperative Blood Testing in Ophthalmic Surgery**
- Led a retrospective study of 102,542 ophthalmic surgical patients, evaluating the necessity of routine preoperative blood testing.
- Evaluated whether Choosing Wisely recommendations could be applied to ophthalmic surgery in China, by integrating epidemiological findings with patient safety and workflow considerations.
- Published the findings as first author in the top journal.

PUBLICATIONS

Peer-reviewed Journal Articles: 6; First-author: 1

1. **Lai, Y.**, Zeng, W., Liao, J., Yu, Y., Liu, X., & Wu, K. (2024) Retrospective analyses of routine preoperative blood testing in a tertiary eye hospital: could Choosing Wisely work in China? *The British journal of ophthalmology*, 108(6), 897–902.
2. Luo, M. J., Pang, J., Bi, S., **Lai, Y.**, et al. (2024) Development and Evaluation of a Retrieval-Augmented Large Language Model Framework for Ophthalmology. *JAMA ophthalmology*, 142(9), 798–805.
3. Luo, M. J., Bi, S., Pang, J., Liu, L., Tsui, C. K., **Lai, Y.**, et al. (2025) A large language model digital patient system enhances ophthalmology history taking skills. *NPJ digital medicine*, 8(1), 502.
4. Liu, L., Hong, J., Wu, Y., ... **Lai, Y.**, et al. (2024) Digital ray: enhancing cataractous fundus images using style transfer generative adversarial networks to improve retinopathy detection. *The British journal of ophthalmology*, 108(10), 1423–1429.
5. Shang, Y., Lin, Y., Li, R., ... **Lai, Y.**, et al. (2025) The effectiveness of large language models in medical AI research for physicians: A randomized controlled trial. *Cell reports. Medicine*, 6(12), 102469.
6. Li, Y., Cui, T., Lin, Z., Zhao, L., **Lai, Y.**, et al. (2026) Risk factors and critical thresholds for glucocorticoid-induced ocular adverse reactions in children: implications for muscle and neuromuscular disease management. *European journal of translational myology*. 36(2).

Submitted Manuscripts

1. Luo, M.J. *, **Lai, Y.***, W, W.*, et al. Standardized Pre-consultation by a Large Language Model Agent vs Ophthalmology Residents: A Randomized Clinical Trial. Revised manuscript in review in *NPJ digital medicine*. (*contributed equally to this study)

TRANSFERABLE SKILLS

Clinical and Microsurgical Training: Ophthalmology and microscope-assisted procedural experience requiring precise visuomotor coordination, sustained attention and careful observation.

Research Design: Experience with retrospective epidemiology studies, prospective longitudinal cohorts and randomized controlled trials.

Project Management and Cross-disciplinary Teamwork: Collaborated across clinical, technical and industry teams to translate clinical needs into research and implementation workflows while completing clinical rotations.

Quantitative and Computational Methods: Statistical analysis using R and SPSS, including regression and mixed-effects modeling; Python programming and AI-assisted coding.

CONFERENCE PRESENTATION

May 2025 **Evaluating GPT-4o in ophthalmic consultation and documentation: Bridging knowledge with clinical practice.** (Poster)
Association for Research in Vision and Ophthalmology (ARVO) Annual Meeting, Salt Lake City, Utah, USA

TEACHING & ACADEMIC SERVICE

Aug 2024 **[Instructor] Pre-Rotation Training for First-Year Ophthalmology Residents**
 - Prepared training materials and delivered practical instruction in essential ophthalmic clinical procedures, including hand hygiene and disinfection.

Jun and Aug 2024 **[Organizing Committee Member] Research Innovation Event**
 - Co-organized the public launch of an ophthalmic LLM, lead-contributed to its visual-identity development, intellectual-property applications and media promoting.
 - Supported institutional activities through video planning and content development.

INTERESTS

Strength training, yoga, and music.